Accelerating Dairy Sustainability: Integrating Economic Analysis into the RuFaS Model for Decision Support

FAB Meeting and Progress Review

Jonah Greene, Braden Limb, and Jason Quinn

Sustainability Science

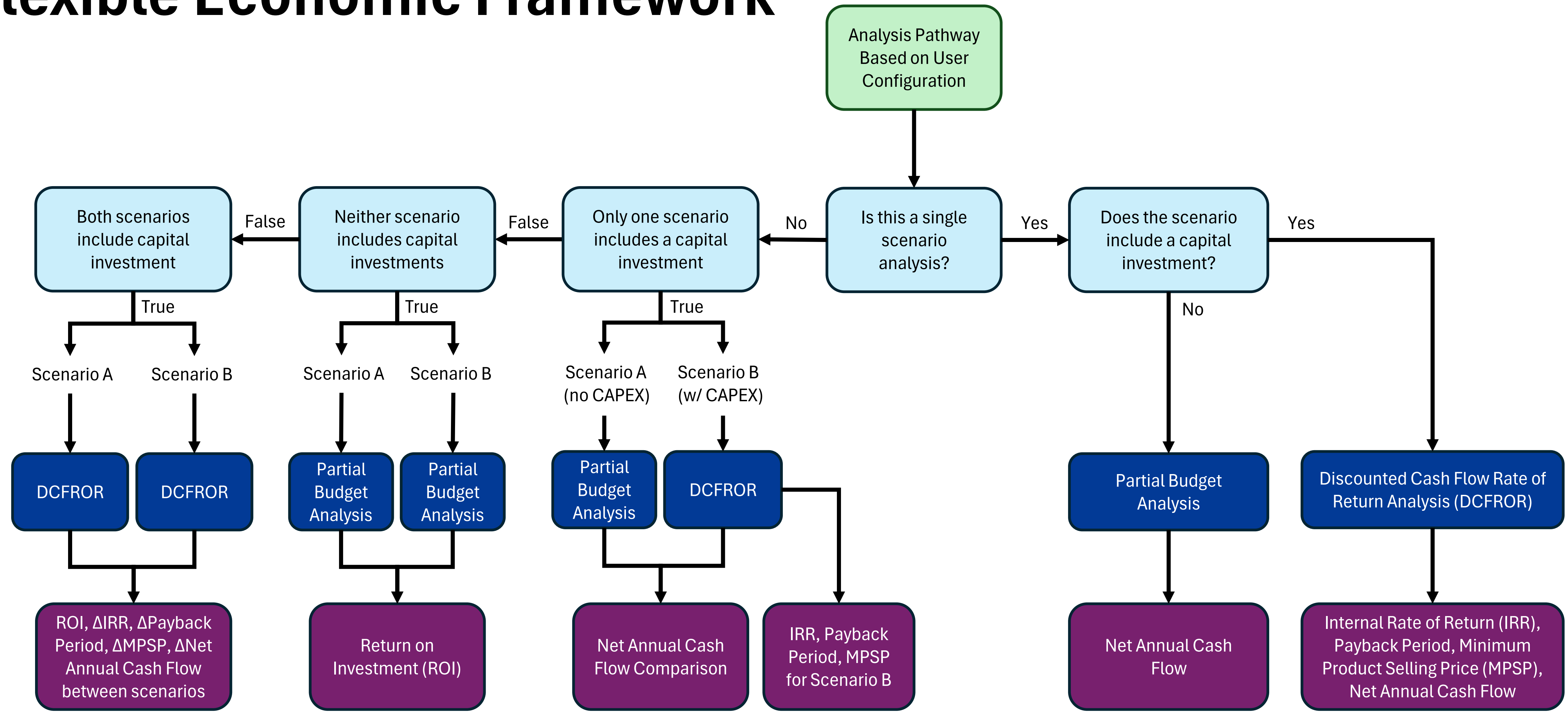
04/25/25

Target Analyses

Target analyses can be analyzed in isolation or combination

- **Beef-on-Dairy Breeding Strategies**: Increase calf value through crossbreeding, balance semen costs with market premiums and replacement needs.
- **High-Quality Feed for Increased Milk Production**: Evaluate returns on feed investments through improved milk output and profitability under varying market conditions.
- **Methane-Reducing Feed Strategies**: Assess feed additives and diet changes for emissions reduction and carbon credit revenue potential.
- **Feed Quality and Manure Nutrient Value**: Link feed improvements to enhanced manure nutrient content and reduced fertilizer costs.
- **Land Allocation for Feed Cost Reduction**: Optimize homegrown feed production to lower purchased feed expenses and maintain quality.
- **Conservation Cropping Practice:** Analyze practices like no-till and cover cropping for soil health, operational savings, and carbon credit opportunities.
- **Manure Management System Investments**: Evaluate storage systems, separators, digesters, and biogas upgrading systems for nutrient management, energy production, and co-product revenue.
- **Animal Comfort Improvements**: Assess barn and ventilation upgrades for impacts on health costs, labor efficiency, and milk yield.
- **Feed Storage Infrastructure Enhancements:** Reduce feed waste and costs by improving storage to maintain feed quality year-round.

Flexible Economic Framework



Sustainability Science

Dynamic Costing Functions

Digester Fixed Costs Equation Inputs:

Total installed capital cost is determined using the annual fixed costs equation from Cowley and Brorsen, then utilizing the capital recovery factor (as a function of discount rate (5%) and equipment lifetime (20 years)) to determine total installed capital cost (in year 1).

Input or Variable Name	Symbol	Default Value (If Applicable)	Units
Function Inputs (from Economic Inputs Manager)			
Digester Population (Animal Units)	A_j	*NA*[1]	AU
Digester Volume	V_j	*NA*[1]	m3
Farm Type (1=Dairy, 0=Otherwise)	F_j	1	Binary
Digester Below Ground (1=Yes, 0=No)	G_j	1	Binary
Primarily Concrete (1=Yes, 0=No)	C_j	1	Binary
Primarily Steel (1=Yes, 0=No)	S_j	0	Binary
Function Constants			
Intercept Term	α	10.4	$/MJ
Construction Cost per AU	R_1	-0.413	$/AU
Construction Cost per Volume	R_2	-0.147	$/m3
Farm Type Constant	ψ_1	-1.085	unitless
Below Ground Constant	ψ_2	-0.582	unitless
Concrete Constant	ψ_3	0.519	unitless
Steel Constant	ψ_4	-0.007	unitless
Standard Error Term	ε_j	0.35	$/MJ

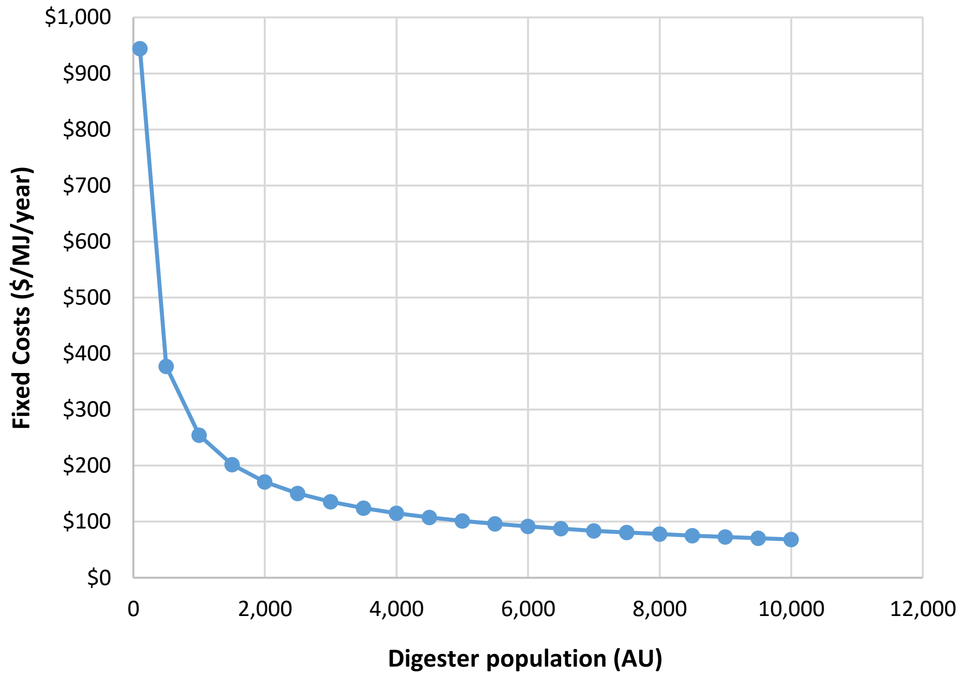
[1]The Economics Module does not offer default values for digester population and volume. These parameters should be tied to the farm size with values provided by the RuFaS biophysical module. However, the user can still override these and supply custom or arbitrary values (for scenario analysis, sensitivity analysis, etc.).

Equations for Digester Capital Costs from Cowley and Brorsen:

$$AFC_{ij} = EXP\left(\alpha + \left(R_1 \times \ln(A_j)\right) + \left(R_2 \times \ln(V_j)\right) + (\psi_1 \times F_j) + (\psi_2 \times G_j) + (\psi_3 \times C_j) + (\psi_4 \times S_j) + \varepsilon_j\right)$$

$$CRF = \frac{r(1+r)^t}{(1+r)^t - 1}$$ (Capital Recovery Factor)

$$CAPEX_{digester} = \frac{AFC_{ij}}{CRF}$$



Dynamic Costing Functions

Other Manure Management Upgrades:

Installed capital costs for other manure management system upgrades are determined using a standardized equipment costing function that applies a scaling relationship between system size and capital cost, along with an installation factor to account for site-specific deployment considerations.

Equipment	Base Flowrate ($V_{m,base}$)	Base Purchase Cost ($C_{p,base}$)	Scaling Exponent (n)	Install Multiplier ($F_{install}$)	Reference
Screw Press Separator	100 gal/min	$75,000	0.65	1.4	USDA NRCS (2018), EPA AgSTAR (2021), Vendor Data
Rotary Screen Separator	100 gal/min	$50,000	0.6	1.4	EPA AgSTAR (2020), USDA NRCS EQIP (2021), Vendor Data
CHP Unit	500 Nm^3/hr	$1,200,000	0.65	1.5	EPA (2021), Vendor Data
RNG Upgrading	500 Nm^3/hr	$1,500,000	0.6	1.6	EPA (2021), IEA (2020), Vendor Data

Standardized Equipment Costing Function:

$$C_{install} = C_{p,base} \times \left(\frac{V_m}{V_{m,base}}\right)^n \times F_{install}$$

Example DCFROR Analysis

Mid-sized dairy installs two 10,000-bushel steel grain bins

- Total Capital Cost: $100,000
- Financing: 80% debt-financed
- Annual OPEX: $5,500 (labor, electricity, fuel, maintenance)
- Savings (Revenue): $40/ton feed cost reduction + 2% less spoilage
- Depreciation: 7-year MACRS schedule
- Target IRR: 10%

Fig. 3 – Sankey Diagram (Cost Breakdown)

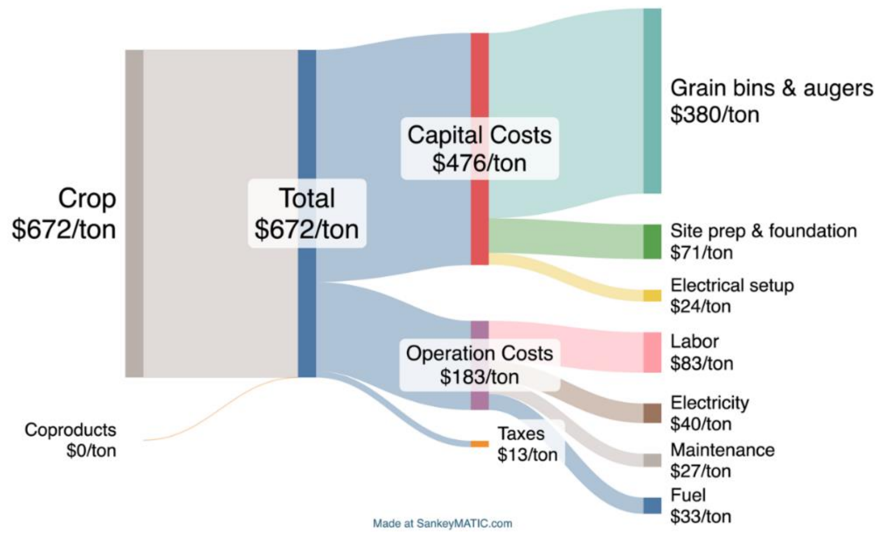
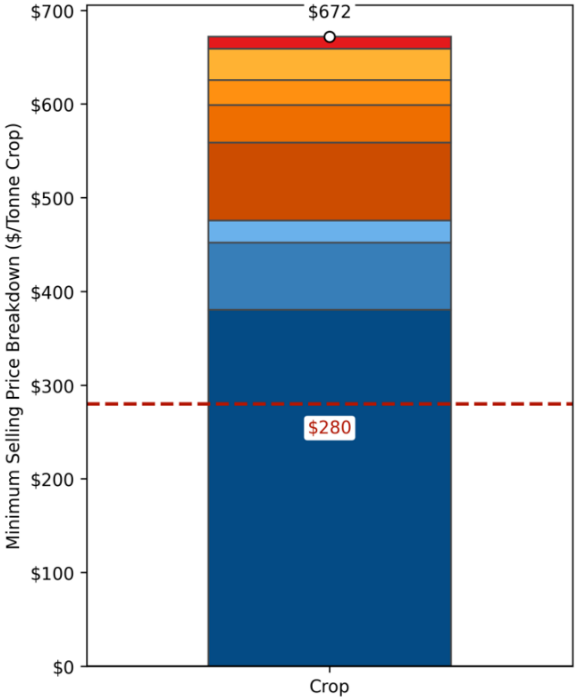


Fig. 2 – Minimum Crop Selling Price



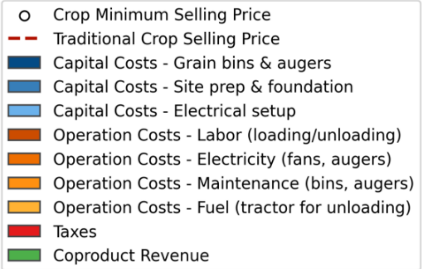
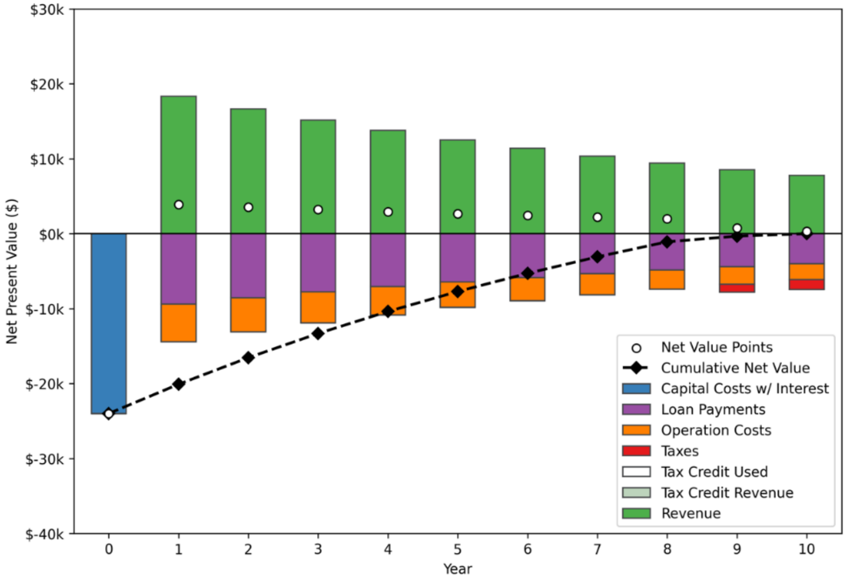


Fig. 1 - Net Present Cashflow Over Project Life

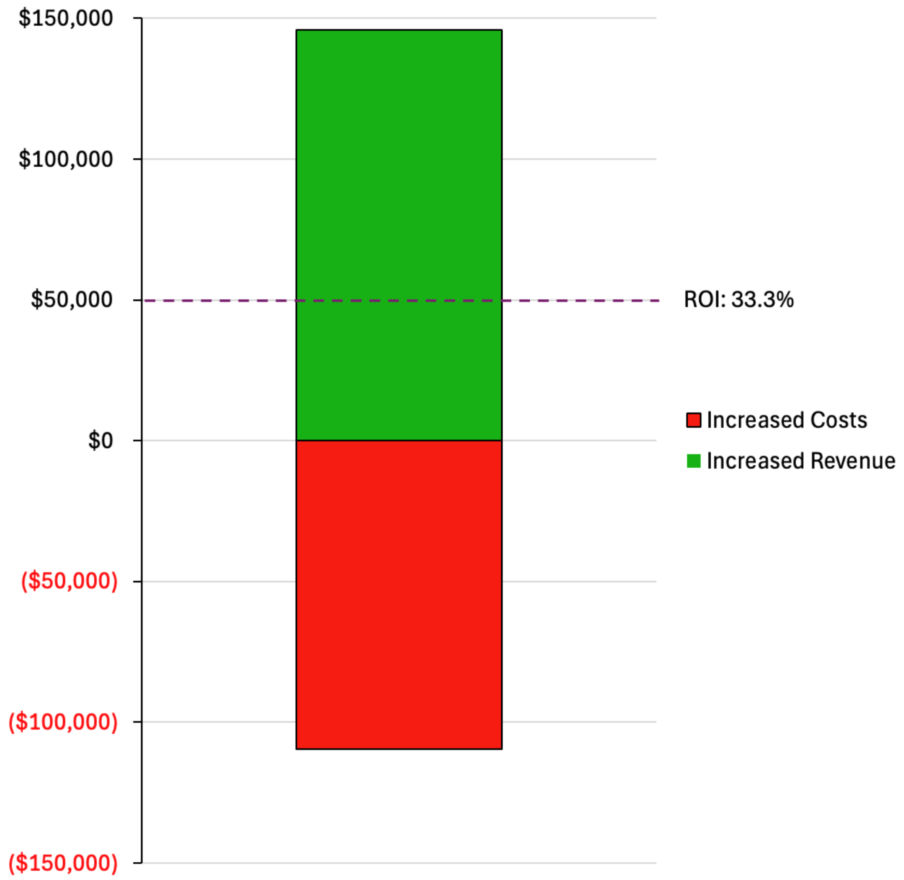


In this example, we find that the maximum feed purchase price or minimum crop selling price is greater than the market feed purchase prices ($652/ton vs $280/ton), shown in Figure 2. **Therefore, it would not be financially profitable to install the feed storage equipment on the farm.**

Example ROI Analysis

A 200-cow dairy farm adopts a more nutrient-dense feed ration in a high milk price environment

Fig. 1 – Costs vs. Revenues and ROI



Assumptions

- Feed Cost Increase: From $0.12/lb to $0.15/lb (with 50 lbs/day per cow)
- Total Additional Cost: $109,500/year
- Milk Output Gain: +5 lbs/day per cow, driven by improved feed quality
- Milk Price Assumption: $0.40/lb (reflecting premiums in organic or value-added markets)

Results

- Revenue Increase: $146,000/year from increased milk production
- Net Profit Gain: $36,500/year.
- Return on Investment (ROI): 33.3%.

Feed optimization can generate significant returns when paired with high-margin milk pricing, emphasizing the importance of evaluating both cost and market value.

Data Gaps and Desired Feedback

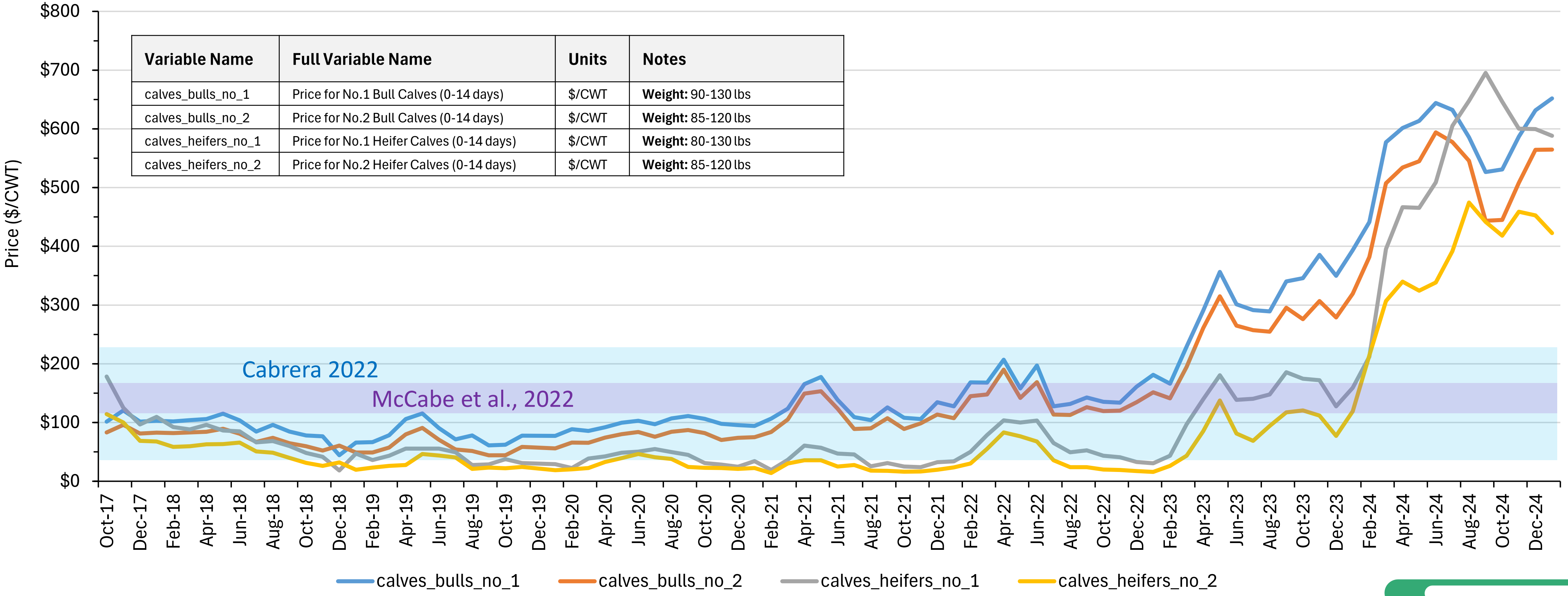
The project team is seeking FAB guidance to close the following data gaps:

1. **Bedding Material Costs:** Regional prices for sand, straw, sawdust, and manure solids.
2. **Semen Prices and Usage:** Current cost trends and use of beef vs. sexed semen across regions.
3. **Carbon Credits:** Relevance, valuation, and markets for carbon credits
4. **Crossbred vs. Dairy Calf Pricing:** Regional prices for Holstein vs. Holstein/beef crossbred calves and cows. While USDA national averages are available, recent spikes in calf value need validation for both dairy and crossbred beef markets. We are also seeking FAB input on whether default values for animal/calf pricing are appropriate or whether they are too variable across time, region, and market type to standardize **(see next slide)**

Cattle Price Data Collected to Date

Temporal Resolution: Monthly (from 10/17 – 1/25)
Spatial Resolution: National Average
Source: USDA MyMarketNews Reports – National Dairy Comprehensive Report

Fig. 1 - Calf Prices (USDA My Market New Reports)



Variable Name	Full Variable Name	Units	Notes
calves_bulls_no_1	Price for No.1 Bull Calves (0-14 days)	$/CWT	**Weight:** 90-130 lbs
calves_bulls_no_2	Price for No.2 Bull Calves (0-14 days)	$/CWT	**Weight:** 85-120 lbs
calves_heifers_no_1	Price for No.1 Heifer Calves (0-14 days)	$/CWT	**Weight:** 80-130 lbs
calves_heifers_no_2	Price for No.2 Heifer Calves (0-14 days)	$/CWT	**Weight:** 85-120 lbs

Thank you for your time and feedback!

Please feel free to reach out with any questions, comments, or concerns!

jonah@sustainabilitysci.com
jason@sustainabilitysci.com
braden@sustainabilitysci.com